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WELLBORE CABLE CLAMP

Abstract

A tubing to be positioned in a wellbore, the tubing comprising: a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore, wherein at least one cable is to be positioned in the channel and comprises at least one of a sensor or a splice; and a clamp configured to be positioned in each of the multiple pairs of slots and configurable to clamp down on the at least one cable positioned in the channel, wherein the clamp is to be positioned in a first pair of slots of the multiple pairs of the slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable.

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Background/Summary

BACKGROUND

[0001] In a wellbore, clamps for holding cables in position may be needed for various applications downhole. For example, a clamp may be needed to hold one or more cables in a channel. One such example is a clamp to hold cables in a channel for a sand control screen. However, often times there may be splices or sensors on these cables that will either be larger than a size of the cables or be sensitive to clamping loads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Embodiments of the disclosure may be better understood by referencing the accompanying drawings.

[0003] FIG. 1 is a perspective view in partial cross section of a wellbore including a pair of sand control screen assemblies having cables, according to some embodiments.

[0004] FIG. 2 is a perspective view of a completion string that includes clamps, according to some embodiments.

[0005] FIG. 3 is a first detailed perspective view of a completion string that includes clamps, according to some embodiments.

[0006] FIG. 4 is another perspective view of a completion string that includes clamps, according to some embodiments.

[0007] FIG. 5 is a second more detailed perspective view of a completion string that includes clamps, according to some embodiments.

[0008] FIG. 6 is a third more detailed perspective view of a completion string that includes clamps, according to some embodiments.

[0009] FIG. 7 is another perspective view of a completion string having multiple sections that include clamps, according to some embodiments.

[0010] FIG. 8 is a more detailed perspective view of a completion string that includes a clamp clamped into position over two cables within a channel, according to some embodiments.

[0011] FIG. 9 is a perspective view of a clamp clamped into position holding down two cables within a channel, according to some embodiments.

[0012] FIGS. 10A-10D are perspective views of an example order of operations for installing cables into a channel and a clamp used for clamping, according to some embodiments.

[0013] FIG. 11 is a perspective view of a channel with cables positioned therein and with the clamp removed, according to some embodiments.

[0014] FIG. 12 is a perspective view of a channel having an example pair of slots cutout from the channel where a clamp may be positioned, according to some embodiments.

[0015] FIG. 13 is a perspective view of a bottom side of a t-slot clamp, according to some embodiments.

[0016] FIG. 14 is a cross-sectional view of clamp positioned for holding two cables in a channel, according to some embodiments.

[0017] FIGS. 15A-15B are perspective views of two different clamps of differing widths, according to some embodiments.

[0018] FIG. 16 is a flowchart of example operations for using a clamp for holding cables in a channel of a tubing for use downhole in a wellbore, according to some embodiments.

DESCRIPTION

[0019] The description that follows includes example systems, methods, techniques, and program

flows that embody aspects of the disclosure. However, it is understood that this disclosure may be practiced without these specific details. In some instances, well-known instruction instances, protocols, structures, and techniques have not been shown in detail in order not to obfuscate the description.

[0020] Example embodiments may include a modular line capture clamp to be used in various downhole wellbore applications. Such a clamp may be used to hold one or more lines in position. In some implementations, this clamp may be used to hold one or more lines in position in a channel over a sand control screen to limit inclusion of solids into the production fluid that is transported to a surface of the wellbore. In particular, during drilling and completion of wellbores, relatively fine particulate materials may be produced during the production of hydrocarbons from a wellbore that traverses an unconsolidated or loosely consolidated formation. Numerous problems may occur as a result of the production of such particulate. For example, the particulate causes abrasive wear to components within the wellbore, such as flow control devices, safety equipment, tubing and the like. In addition, the particulate may partially or fully clog the wellbore creating the need for an expensive workover. Also, if the particulate matter is produced to the surface, it must be removed from the hydrocarbon fluids using surface processing equipment. Additionally, problems may occur downhole if the cables running through the sand control screens are hanging loose (instead of being clamped down). For example, the cables may get hung on different equipment downhole while being positioned downhole (thereby causing damage to such cables). For example, if the cables are fiber optic or electrical cables (or include sensors), such damage to the cables may result in inoperability of these cables. These cables may include splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. These splices or sensors can land anywhere, so a clamp that can be moved easily is needed.

[0021] Example implementations may include a modular type clamp to keep the cables at an essentially fixed position. In some implementations, the cables may be run along a channel and the channel may include multiple slots such that a given clamp may be easily moved to a different slot along the channel to avoid clamping down on the cable where a sensor and/or splice may be.

[0022] Example of the cables may be fiber optic cables, electrical cables, hydraulic cables, etc. The clamps may provide a level of compression on the cables. Accordingly, the clamps may provide some amount of support for the cable-so that the device that the cable is attached to does not have to support all of the weight of the cable between connection points.

[0023] Accordingly, example implementations may include a number of different slots in the channel at different depths of the wellbore so that if a sensor or splice is at one location of a pair of slots, the clamp may be moved to a different pair of slots. In some implementations, the clamp may be wider to provide more protection for the cables. For example, a width of the clamp may be at least 3 inches. However, in such implementations, the clamp may not be rotatable-thereby requiring the clamp to be removed from the channel in order to remove the cables from the channel.

[0024] In some implementations, the clamp may a simple t-slot type clamp. Since cutting (using laser, stamping, water jet, etc.) a sheet metal channel may be quick and easy, there may be many locations for these slot locations. Having this large number of mounting options allows for quick and easy modularity of the placement of these clamps. Some implementations may include at least 2× more slot locations than clamps provided. Accordingly, if a sensor is positioned on a cable where a clamp is also positioned, that clamp may be removed and relocated to the nearest slot location. In some implementations, the clamp does not need to be removed from the joint in order to install the cables. Rather, the clamp may just need to be loosened and rotated.

[0025] Accordingly, example implementations may include a T-slot clamp mounting design that may allow for modularity of clamping locations. Additionally, example implementations may include slots in a channel that the clamp may be dropped into after tightening that provide additional security for tensile and/or slippage of the cables.

[0026] Therefore, example implementations may provide a solution for handling the problem of

unknown locations of sensors and splice landings in a cable positioned downhole in a wellbore. Also, example implementations may be tailored to a variety of cable designs by a simple clamp swap out (without changing the channel that may already be in stock with a customer).

Additionally, in some implementations, a clamp lid may be customized for a changing number of cables or shapes and sizes. Also, it may be possible to change the clamp during the running of the system if cable configurations are changed at the last minute due to unforeseen issues.

[0027] While depicted such that cables are not essentially completely covered by such clamps, some implementations may be configured to provide for such covering. For example, clamps may be widened and placed adjacent to each other so that the cables are essentially completely covered. In some such implementations, a given clamp may be removed if a sensor or splice is in the cable at that location.

[0028] Accordingly, example implementations may include a modular t-slot clamp that may be removed or relocated as necessary to avoid clamping over a sensor or splice of the cable. The clamp may be considered modular because the clamp may be positioned at one of any number of locations in the channel where a pair of slots are positioned.

[0029] While described such that the channel with the multiple pair of slots along with modular cable clamps are part of a protective shroud around sand control screens as part of a completion string, example implementations may be used in other configurations. For example, the channel with the multiple pair of slots along with modular cable clamps may be integrated into the sand control screens themselves. In other examples, the channel with the multiple pair of slots along with modular cable clamps may be part of a downhole tubing that does not enclose sand control screens. Further, the channel with the multiple pair of slots along with modular cable clamps may be integrated into any type of downhole tubing (for production, completion, drilling, etc.).

[0030] Additionally, in some implementations, a clamp may be positioned on each side of a sensor and a covering or housing may be added between to cover or protect the sensor. In some implementations, the bottom part (orange) may be on each side of a sensor and a top part of the clamp may extend between the two bottom parts to provide a covering for the sensor.

Example Wellbore System

[0031] An example wellbore system having modular cable clamps is now described. FIG. 1 is a perspective view in partial cross section of a wellbore including a pair of sand control screen assemblies having cables, according to some embodiments. In FIG. 1, a wellbore 12 extends through the various earth strata including formations 14, 16. A casing 18 is supported within wellbore 12 by cement 20. A completion string 22 includes various tools such as a sand control screen assembly 24 that is positioned within production interval 26 between packers 28, 30. In addition, completion string includes a sand control screen assembly 32 that is positioned within production interval 34 between packers 36, 38. One or more cables 40 extend from the surface within annulus 42 as pass through sand control screen assemblies 24, 32 to provide instructions, carry power, signals and data, and transport operating fluid, such as hydraulic fluid, to sensors, actuators and the like associated with sand control screen assemblies 24, 32 and other tools or components positioned downhole. In some implementations, the cables 40 may be fiber optic cables, electrical cables, hydraulic cables, etc.

[0032] In one example, once completion string 22 is positioned as shown within wellbore 12, a treatment fluid containing sand, gravel, proppants or the like may be pumped down completion string 22 such that formations 14, 16 and production intervals 26, 34 may be treated. Sensors operably associated with completion string 22 may be used to provide substantially real time data to the operator via the cable 40 on the effectiveness of the treatment operation such as identifying voids during the gravel placement process to allow the operator to adjust treatment parameters such as pump rate, proppant concentration, fluid viscosity and the like to overcome deficiencies in the gravel pack. In addition, such sensors may be used to provide valuable information to the operator via cable 40 during the production phase of the well such as fluid temperature, pressure, velocity,

constituent composition and the like such that the operator can enhance the production operations. [0033] Even though FIG. 1 depicts sand control screen assemblies **24, 32** in a cased hole environment, it should be understood by those skilled in the art that the sand control screen assemblies of the present invention are equally well suited for use in open hole environments. Also, even though FIG. 1 depicts a single sand control screen assembly having three screen jackets in each production interval, it should be understood by those skilled in the art that any number of sand control screen assemblies each having any number of screen jackets may be deployed within a production interval without departing from the principles of the present invention. Further, even though FIG. 1 depicts a vertical completion, it should be understood by those skilled in the art that the sand control screen assemblies of the present invention are equally well suited for use in well having other directional configurations including horizontal wells, deviated wells, slanted wells, multilateral wells and the like. Accordingly, it should be understood by those skilled in the art that the use of directional terms such as above, below, upper, lower, upward, downward, left, right, uphole, downhole and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward direction being toward the top of the corresponding figure and the downward direction being toward the bottom of the corresponding figure, the uphole direction being toward the surface of the well and the downhole direction being toward the toe of the well.

Example Completion Strings

[0034] Example completion string having modular cable clamps are now described. FIG. 2 is a perspective view of a completion string that includes clamps, according to some embodiments. In particular, FIG. 2 depicts a completion string **200** that may be representative of the completion string **22** of FIG. 1. In some implementations, a diameter of the completion string **200** may be smaller than a diameter of the wellbore so that an annulus may be formed between the outer wall of the completion string and the wall of the wellbore.

[0035] In this example, the completion string **200** includes a protective shroud **202** that encloses a sand control screen assembly **204**. The protective shroud **202** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **204**. The protective shroud **202** also includes a channel **206**. Cables **208-210** are run within the channel **206**. Additionally, cable clamps **220-230** may be modular so that the cable clamps **220-230** may be positioned at different locations along the channel **206** to clamp down and secure the cables **208-210** in the channel **206**.

[0036] In some implementations, the channel **206** may include multiple pairs of slots into which a given cable clamp is positioned. Additionally, cable clamps **220-230** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamps **220-230** may be moved to different pairs of the slots so that the cable clamps **220-230** are not clamped onto the sensors or splices.

[0037] FIG. 3 is a first detailed perspective view of a completion string that includes clamps, according to some embodiments. In particular, FIG. 3 depicts a completion string **300** that may be representative of the completion string **22** of FIG. 1. In this example, the completion string **300** includes a protective shroud **302** that encloses a sand control screen assembly **304**. The protective shroud **302** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **304**. The protective shroud **302** also includes a channel **306**. Cables **308-310** are run within the channel **306**. Additionally, cable clamps **320-330** may be modular so that the cable clamps **320-330** may be positioned at different locations along the channel **306** to clamp down and secure the cables **308-310** in the channel **306**.

[0038] The channel **306** includes multiple pairs of slots **350-370** into which a given cable clamp is positioned. Additionally, the cables **408-410** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamps **320-330** may be moved to different pairs of the slots so that the cable clamps **320-330** are not clamped onto the sensors or splices.

[0039] FIG. 4 is another perspective view of a completion string that includes clamps, according to some embodiments. In particular, FIG. 4 depicts a completion string **400** that may be representative of the completion string **22** of FIG. 1. In this example, the completion string **400** includes a protective shroud **402** that encloses a sand control screen assembly. The protective shroud **402** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly. The protective shroud **402** also includes a channel **406**. Cables **408-410** are run within the channel **406**. Additionally, cable clamps **420-422** may be modular so that the cable clamps may be positioned at different locations along the channel **406** to clamp down and secure the cables **408-410** in the channel **406**.

[0040] The channel **406** includes multiple pairs of slots into which a given cable clamp is positioned. Additionally, the cables **408-410** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamps **420-422** may be moved to different pairs of the slots so that the cable clamps **420-422** are not clamped onto the sensors or splices.

[0041] FIG. 5 is a second more detailed perspective view of a completion string that includes clamps, according to some embodiments. In particular, FIG. 5 depicts a completion string **500** that may be representative of the completion string **22** of FIG. 1. In this example, the completion string **500** includes a protective shroud **502** that encloses a sand control screen assembly **504**. The protective shroud **502** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **504**. The protective shroud **502** also includes a channel **506**. Cables **508-510** are run within the channel **506**. Additionally, a cable clamp **520** may be modular so that the cable clamp **520** may be positioned at different locations along the channel **506** to clamp down and secure the cables **508-510** in the channel **506**.

[0042] The channel **506** includes multiple pairs of slots into which a given cable clamp is positioned. Additionally, the cables **508-510** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamp **520** may be moved to different pairs of the slots so that the cable clamp **520** is not clamped onto the sensors or splices.

[0043] FIG. 6 is a third more detailed perspective view of a completion string that includes clamps, according to some embodiments. In particular, FIG. 6 depicts a completion string **600** that may be representative of the completion string **22** of FIG. 1. In this example, the completion string **600** includes a protective shroud **602** that encloses a sand control screen assembly **604**. The protective shroud **602** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **604**. The protective shroud **602** also includes a channel **606**. Cables **608-610** are run within the channel **606**. Additionally, a cable clamp **620** may be modular so that the cable clamp **620** may be positioned at different locations along the channel **606** to clamp down and secure the cables **608-610** in the channel **606**.

[0044] The channel **606** includes multiple pairs of slots into which a given cable clamp is positioned. Additionally, the cables **608-610** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamp may be moved to different pairs of the slots so that the cable clamp **620** are not clamped onto the sensors or splices.

[0045] FIG. 7 is another perspective view of a completion string having multiple sections that include clamps, according to some embodiments. In particular, FIG. 7 depicts a completion string **700** that may be representative of the completion string **22** of FIG. 1. In this example, the completion string **700** includes a protective shroud **702** that encloses a sand control screen assembly **704**. The protective shroud **702** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **704**. The protective shroud **702** also includes a channel **706**. Cables **708-710** are run within the channel **706**.

Additionally, cable clamps **720-740** may be modular so that the cable clamps **720-740** may be

positioned at different locations along the channel **706** to clamp down and secure the cables **708-710** in the channel **706**.

[0046] The channel **706** includes multiple pairs of slots into which a given cable clamp is positioned. Additionally, the cables **708-710** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamps **720-740** may be moved to different pairs of the slots so that the cable clamps **720-740** are not clamped onto the sensors or splices.

[0047] FIG. **8** is a more detailed perspective view of a completion string that includes a clamp clamped into position over two cables within a channel, according to some embodiments. In particular, FIG. **8** depicts a completion string **800** that may be representative of the completion string **22** of FIG. **1**. In this example, the completion string **800** includes a protective shroud **802** that encloses a sand control screen assembly **804**. The protective shroud **802** may include a number of perforations to allow for formation fluid to flow there through for filtering by the sand control assembly **804**. The protective shroud **802** also includes a channel **806**. A cables **810** is run within the channel **806**. Additionally, a cable clamps **820** may be modular so that the cable clamps **820** may be positioned at different locations along the channel **806** to clamp down and secure the cables **810** in the channel **806**.

[0048] The channel **806** includes multiple pairs of slots into which a given cable clamp is positioned. Additionally, the cables **810** may include sensors or splices. The splices or sensors that may either be larger than a size of the cables or be sensitive to clamping loads. Accordingly, the cable clamp **820** may be moved to different pairs of the slots so that the cable clamp **820** are not clamped onto the sensors or splices.

Example Clamps

[0049] FIG. **9** is a perspective view of a clamp clamped into position holding down two cables within a channel, according to some embodiments. FIG. **9** depicts a channel **906** that includes a pair of slots **950**. FIG. **9** also depicts cables **908-910** positioned in the channel **906**. Additionally, FIG. **9** depicts a clamp **920** positioned in the pair of slots **950**. The clamp **950** may include a rotatable top **981**, a bottom anchor **980**, and a screw **982**.

[0050] FIGS. **10A-10D** are perspective views of an example order of operations for installing cables into a channel and a clamp used for clamping, according to some embodiments. FIG. **10A** depicts a clamp **1020** in a non-clamped position without cables being positioned in a channel **1006**. The clamp **1020** includes a rotatable top **1081**, a bottom anchor **1080**, and a screw **1082**. The channel **1006** includes a pair of slots **1050** into which the clamp **1020** may be positioned.

[0051] For example, the channel **1006** may be part of a tubing to be positioned downhole in a wellbore. For instance, after the tubing is brought to a well rig site to be positioned downhole in the wellbore, the clamp **1020** may be rotated such that the clamp **1020** is essentially parallel with the longitudinal axis of the channel **1006**.

[0052] As shown, in FIG. **10B**, after the rotatable top **1081** is rotated, cables **1008-1010** may then be placed in the channel **1006** and slid past the clamp **1020** through the channel **1006** in order to position the cables **1009-1010** in the channel **1006** prior to positioning the tubing downhole in the wellbore. In FIG. **10C**, the rotatable top **1081** of the clamp **1020** is rotated back to a clamping position above the pair of slots **1050** in the channel **1006**. In FIG. **10D**, the clamp **1020** may then be clamped down in the pair of slots **1050** and tightened therein using the screw **1082** of the clamp **1020**. Additionally, as shown, after the clamp **1020** is positioned in the pair of slots **1050** and tightened down, the pair of slots **1050** may preclude the clamp **1020** from slipping axially. Accordingly, as shown, there is no requirement that the clamp **1020** be removed in order to move and/or remove the cables **1008-1010** from the channel **1006**.

[0053] In some implementations, a length of the clamp **1020** is such that the clamp **1020** may be rotated (without being removed from the channel **1006**) in order to move or remove the cables **1008-1010** from the channel **1006**. Accordingly, in some implementations, a length of the clamp

1020 may be directly correlated with a width of the channel **1006** in order to allow for rotation of the clamp **1020** without removal of the clamp **1020** from the channel **1006** in order to allow for movement or removal of the cables **1008-1010** from the channel **1006**. Thus, the wider the channel **1006**, the longer the clamp **1020** may be. In some implementations, a length of clamp **1020** across the channel **1006** is less than a width of the channel **1006** but long enough to remain in position within the pair of slots **1050**.

[0054] FIG. **11** is a perspective view of a channel with cables positioned therein and with the clamp removed, according to some embodiments. FIG. **11** depicts a channel **1106** with the clamped removed. The channel **1106** also includes a pair of slots **1150**. FIG. **11** also depicts cables **1108-1110** being positioned in the channel **1106**. Removing the clamp allows for additional space for sensors or splices or areas in the cables more sensitive to clamping.

[0055] FIG. **12** is a perspective view of a channel having an example pair of slots cutout from the channel where a clamp may be positioned, according to some embodiments. FIG. **12** depicts a channel **1206** that includes a pair of slots **1250** into which a clamp may be positioned for holding cables in the channel **1206**.

[0056] FIG. **13** is a perspective view of a bottom side of a t-slot clamp, according to some embodiments. FIG. **13** depicts a clamp **1320** that includes a rotatable top **1381**, a bottom anchor **1380** on which the rotatable top **1381** rotates, and a screw **1382** that is threaded through the rotatable top **1381** and the bottom anchor **1380**. A bottom side of the rotatable top **1381** includes a recessed area **1394** to form a non-recessed area **1390** and a non-recessed area **1391** on each side. The bottom side of the rotatable top **1381** also includes a recessed area **1395** to also form a non-recessed area **1392** and a non-recessed area **1393** on each side. These non-recessed areas **1390-1391** and **1392-1393** allow the clamp to increase the pressure on the cables as the clamp is tightened down onto the cables. As the clamp is tightened down, the recessed areas **1394-1395** may serve as locations where the covering of the cable may move into. Accordingly, the recessed areas **1390-1393** may allow the clamp to increase the bite into the encapsulation of the cables. Such implementations allow the clamp to support at least a portion of the weight of the cable. In some implementations, there may be a hole **1396** in the threaded area of the bottom anchor **1380**. After the screw **1382** is tightened into the threaded area, a pin may be placed into the hole **1396** to deform the threads, the pin and threads may be struck to deform the threads so that the screw **1382** does not loosen.

[0057] FIG. **14** is a cross-sectional view of clamp positioned for holding two cables in a channel, according to some embodiments. FIG. **14** depicts a clamp **1420** that includes a rotatable top **1402**, a bottom anchor **1404**, and a screw **1482**. FIG. **14** also depicts a channel **1406** and cables **1408-1410** positioned therein. The clamp **1420** is positioned above the channel **1406** to clamp the cables **1408-1410** down into a position in the channel **1406**.

[0058] FIGS. **15A-15B** are perspective views of two different clamps of differing widths, according to some embodiments. FIGS. **15A-15B** depict a clamp **1502** and a clamp **1504**, respectively. As shown, the clamp **1502** is wider than the clamp **1504**. Also, a width of the clamp **1502** is wider than the channel. Accordingly, in order to remove the cables, the clamp **1502** would need to be removed from the channel.

Example Operations

[0059] Example operations for using a clamp are now described. In particular, FIG. **16** is a flowchart of example operations for using a clamp for holding cables in a channel of a tubing for use downhole in a wellbore, according to some embodiments. Operations of a flowchart **1600** start at block **1602**.

[0060] At block **1602**, a tubing that is to be positioned downhole in a wellbore is received. The tubing comprises a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore.

[0061] At block **1604**, a cable clamp positioned in a first pair of slots of the multiple pair of slots is

loosened.

[0062] At block **1606**, a rotatable top of the cable clamp is rotated such that an axis of the rotatable top is substantially parallel with an axis of the channel.

[0063] At block **1608**, at least one cable is positioned in the channel below the rotatable top of the cable clamp. The at least one cable comprises at least one of a sensor or a splice, wherein the cable clamp is to be positioned in the first pair of slots of the multiple pairs of slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable.

[0064] At block **1610**, the rotatable top of the cable clamp is rotated such that the axis of the rotatable top is substantially perpendicular with the axis of the channel.

[0065] At block **1612**, the cable clamp is tightened into the pair of slots to secure the at least one cable in the channel.

[0066] While the aspects of the disclosure are described with reference to various implementations and exploitations, it will be understood that these aspects are illustrative and that the scope of the claims is not limited to them. In general, techniques for simulating drill bit abrasive wear and damage during the drilling of a wellbore as described herein may be implemented with facilities consistent with any hardware system or hardware systems. Many variations, modifications, additions, and improvements are possible.

[0067] Plural instances may be provided for components, operations or structures described herein as a single instance. Finally, boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the disclosure. In general, structures and functionality presented as separate components in the example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements may fall within the scope of the disclosure.

[0068] The flowcharts are provided to aid in understanding the illustrations and are not to be used to limit scope of the claims. The flowcharts depict example operations that can vary within the scope of the claims. Additional operations may be performed; fewer operations may be performed; the operations may be performed in parallel; and the operations may be performed in a different order. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by program code. The program code may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable machine or apparatus.

[0069] Use of the phrase “at least one of” preceding a list with the conjunction “and” should not be treated as an exclusive list and should not be construed as a list of categories with one item from each category, unless specifically stated otherwise. A clause that recites “at least one of A, B, and C” can be infringed with only one of the listed items, multiple of the listed items, and one or more of the items in the list and another item not listed.

[0070] As used herein, the term “or” is inclusive unless otherwise explicitly noted. Thus, the phrase “at least one of A, B, or C” is satisfied by any element from the set {A, B, C} or any combination thereof, including multiples of any element.

EXAMPLE EMBODIMENTS

[0071] Embodiment #1: A tubing to be positioned in a wellbore, the tubing comprising: a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore, wherein at least one cable is to be positioned in the channel and comprises at least one of a sensor or a splice; and a clamp configured to be positioned in each of the multiple pairs of slots and configurable to clamp down on the at least one cable positioned in the channel, wherein the clamp is to be positioned in a first pair of slots of the multiple pairs of the slots such that clamp is not positioned over the at least one sensor

or the splice of the least one cable. [0072] Embodiment #2: The tubing of Embodiment #1, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable. [0073] Embodiment #3: The tubing of Embodiment #1, wherein the clamp comprises a t-slot clamp. [0074] Embodiment #4: The tubing of Embodiment #3, wherein the clamp comprises a screw to tighten the clamp in position in the first pair of slots in the channel. [0075] Embodiment #5: The tubing of Embodiment #4, wherein the clamp is configured to move between the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore. [0076] Embodiment #6: The tubing of Embodiment #5, wherein the clamp configured to move between the first pair of slots in the channel to the second pair of slots comprises the screw of the clamp configured to loosen, the clamp configured to rotate such that the clamp is not within the first pair of slots, the clamp configured to move to a different depth of the wellbore associated with the second pair of slots, the clamp configured to rotate back to enable the clamp to be positioned in the second pair of slots, and the screw of the clamp configured to retighten. [0077] Embodiment #7: The tubing of Embodiment #1, wherein a length of the clamp is less than a width of the channel. [0078] Embodiment #8: The tubing of Embodiment #1, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing through the sand control screen from a surrounding subsurface formation into which the wellbore is formed. [0079] Embodiment #9: An apparatus comprising: a cable clamp configurable to be positioned in each of the multiple pairs of slots within a channel that is part of a tubing to be positioned downhole in a wellbore, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore, wherein at least one cable is to be positioned in the channel and comprises at least one of a sensor or a splice, wherein the clamp is to be positioned in a first pair of slots of the multiple pairs of the slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable. [0080] Embodiment #10: The apparatus of Embodiment #9, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable. [0081] Embodiment #11: The apparatus of Embodiment #9, wherein the cable clamp comprises a t-slot clamp. [0082] Embodiment #12: The apparatus of Embodiment #11, wherein the cable clamp comprises a screw to tighten the cable clamp in position in the first pair of slots in the channel. [0083] Embodiment #13: The apparatus of Embodiment #12, wherein the cable clamp is configured to move between the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore. [0084] Embodiment #14: The apparatus of Embodiment #13, wherein the cable clamp configured to move between the first pair of slots in the channel to the second pair of slots comprises the screw of the cable clamp configured to loosen, the cable clamp configured to rotate such that the cable clamp is not within the first pair of slots, the cable clamp configured to move to a different depth of the wellbore associated with the second pair of slots, the cable clamp configured to rotate back to enable the cable clamp to be positioned in the second pair of slots, and the screw of the cable clamp configured to retighten. [0085] Embodiment #15: The apparatus of Embodiment #9, wherein a length of the cable clamp is less than a width of the channel. [0086] Embodiment #16: The apparatus of Embodiment #9, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing through the sand control screen from a surrounding subsurface formation into which the wellbore is formed. [0087] Embodiment #17: A method comprising: receiving a tubing that is to be positioned downhole in a wellbore, wherein the tubing comprises a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore; loosening a cable clamp positioned in a first pair of slots of the multiple pair of slots; rotating a rotatable top of the cable clamp such that an axis of the rotatable top is substantially parallel with an axis of the channel; positioning at least one cable in the channel below the rotatable top of the cable clamp, wherein the at least one cable comprises at least one of a sensor or a splice, wherein

the cable clamp is to be positioned in the first pair of slots of the multiple pairs of slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable; rotating the rotatable top of the cable clamp such that the axis of the rotatable top is substantially perpendicular with the axis of the channel; and tightening the cable clamp into the pair of slots to secure the at least one cable in the channel. [0088] Embodiment #18: The method of Embodiment #17, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable. [0089] Embodiment #19: The method of Embodiment #17, wherein the cable clamp comprises a t-slot clamp. [0090] Embodiment #20: The method of Embodiment #19, wherein the cable clamp comprises a screw to tighten the clamp in position in the first pair of slots in the channel. [0091] Embodiment #21: The method of Embodiment #20, further comprising: [0092] moving the cable clamp from the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore. [0093] Embodiment #22: The method of Embodiment #21, moving the cable clamp comprises: loosening the screw of the cable clamp; rotating the cable clamp such that the cable clamp is not within the first pair of slots; moving the cable clamp to a different depth of the wellbore associated with the second pair of slots; rotating the cable clamp back to enable the cable clamp to be positioned in the second pair of slots, and retightening the screw of the clamp. [0094] Embodiment #23: The method of Embodiment #17, wherein a length of the clamp is less than a width of the channel. [0095] Embodiment #24: The method of Embodiment #17, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing through the sand control screen from a surrounding subsurface formation into which the wellbore is formed.

Claims

1. A tubing to be positioned in a wellbore, the tubing comprising: a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore, wherein at least one cable is to be positioned in the channel and comprises at least one of a sensor or a splice; and a clamp configured to be positioned in each of the multiple pairs of slots and configurable to clamp down on the at least one cable positioned in the channel, wherein the clamp is to be positioned in a first pair of slots of the multiple pairs of the slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable.
2. The tubing of claim 1, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable.
3. The tubing of claim 1, wherein the clamp comprises a t-slot clamp.
4. The tubing of claim 3, wherein the clamp comprises a screw to tighten the clamp in position in the first pair of slots in the channel.
5. The tubing of claim 4, wherein the clamp is configured to move between the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore.
6. The tubing of claim 5, wherein the clamp configured to move between the first pair of slots in the channel to the second pair of slots comprises the screw of the clamp configured to loosen, the clamp configured to rotate such that the clamp is not within the first pair of slots, the clamp configured to move to a different depth of the wellbore associated with the second pair of slots, the clamp configured to rotate back to enable the clamp to be positioned in the second pair of slots, and the screw of the clamp configured to retighten.
7. The tubing of claim 1, wherein a length of the clamp is less than a width of the channel.
8. The tubing of claim 1, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing

through the sand control screen from a surrounding subsurface formation into which the wellbore is formed.

9. An apparatus comprising: a cable clamp configurable to be positioned in each of the multiple pairs of slots within a channel that is part of a tubing to be positioned downhole in a wellbore, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore, wherein at least one cable is to be positioned in the channel and comprises at least one of a sensor or a splice, wherein the clamp is to be positioned in a first pair of slots of the multiple pairs of the slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable.

10. The apparatus of claim 9, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable.

11. The apparatus of claim 9, wherein the cable clamp comprises a t-slot clamp.

12. The apparatus of claim 11, wherein the cable clamp comprises a screw to tighten the cable clamp in position in the first pair of slots in the channel.

13. The apparatus of claim 12, wherein the cable clamp is configured to move between the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore.

14. The apparatus of claim 13, wherein the cable clamp configured to move between the first pair of slots in the channel to the second pair of slots comprises the screw of the cable clamp configured to loosen, the cable clamp configured to rotate such that the cable clamp is not within the first pair of slots, the cable clamp configured to move to a different depth of the wellbore associated with the second pair of slots, the cable clamp configured to rotate back to enable the cable clamp to be positioned in the second pair of slots, and the screw of the cable clamp configured to retighten.

15. The apparatus of claim 9, wherein a length of the cable clamp is less than a width of the channel.

16. The apparatus of claim 9, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing through the sand control screen from a surrounding subsurface formation into which the wellbore is formed.

17. A method comprising: receiving a tubing that is to be positioned downhole in a wellbore, wherein the tubing comprises a channel having multiple pairs of slots created on each side of the channel, wherein each of the multiple pairs of slots is positioned at different depths of the wellbore; loosening a cable clamp positioned in a first pair of slots of the multiple pair of slots; rotating a rotatable top of the cable clamp such that an axis of the rotatable top is substantially parallel with an axis of the channel; positioning at least one cable in the channel below the rotatable top of the cable clamp, wherein the at least one cable comprises at least one of a sensor or a splice, wherein the cable clamp is to be positioned in the first pair of slots of the multiple pairs of slots such that clamp is not positioned over the at least one sensor or the splice of the least one cable; rotating the rotatable top of the cable clamp such that the axis of the rotatable top is substantially perpendicular with the axis of the channel; and tightening the cable clamp into the pair of slots to secure the at least one cable in the channel.

18. The method of claim 17, wherein the at least one cable comprises at least one of a fiber optic cable, an electrical cable, or a hydraulic cable.

19. The method of claim 17, wherein the cable clamp comprises a t-slot clamp.

20. The method of claim 19, wherein the cable clamp comprises a screw to tighten the clamp in position in the first pair of slots in the channel.

21. The method of claim 20, further comprising: moving the cable clamp from the first pair of slots in the channel to a second pair of slots of the multiple pairs of slots at a different depth of the wellbore.

22. The method of claim 21, moving the cable clamp comprises: loosening the screw of the cable clamp; rotating the cable clamp such that the cable clamp is not within the first pair of slots;

moving the cable clamp to a different depth of the wellbore associated with the second pair of slots; rotating the cable clamp back to enable the cable clamp to be positioned in the second pair of slots, and retightening the screw of the clamp.

23. The method of claim 17, wherein a length of the clamp is less than a width of the channel.

24. The method of claim 17, wherein the channel is formed in a perforated shroud that is to enclose a sand control screen that is to limit particulates entering therethrough from a formation fluid flowing through the sand control screen from a surrounding subsurface formation into which the wellbore is formed.
